Structure-Compatible Generalization

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Predicting OOD behavior from preserved transformation structure

Abstract

Modern learning systems are underspecified: many finite models can fit the same training data and pass in-distribution checks while behaving differently under deployment shifts. This paper reports a controlled diagnostic suite testing whether the transformation structure a learned function preserves predicts OOD behavior better than loss, validation, norm, or sharpness proxies. The suite combines cyclic symbolic MLPs, rotated visual objects, and a modular algorithmic table task under one common schema. The claim is bounded: for finite domains where deployment shift is generated by a known transformation family, compatibility is an OOD diagnostic and model-selection signal, not a universal certification theorem.

1. Underspecification Target

The practical question is which model to trust when train loss and ID validation do not identify the transportable rule. The diagnostic tested here measures compatibility with transformations expected to generate deployment cases, then compares that signal with standard selection proxies.

2. Suite

The suite reuses the cyclic symbolic and rotated vision weakness benchmarks and adds a modular addition task. Each model emits one common row: train accuracy, ID validation accuracy, OOD accuracy, true compatibility, wrong-group compatibility, loss, parameter norm, and sharpness when available.

Field	Value
domains	symbolic, vision, modular
GPU	L4
symbolic models	128
vision models	96
modular models	128
rows	354

Table 1. Suite manifest.

3. Predictor Rankings

Domain	Top predictor	Pearson r	Mean OOD
modular_exact	compatibility_inferred	1.000	0.545
modular_neural	compatibility_true	0.649	0.165
symbolic_cyclic	compatibility_true	0.763	0.394
vision_rotation	compatibility_true	0.451	0.380

Table 2. Top OOD predictors by domain.

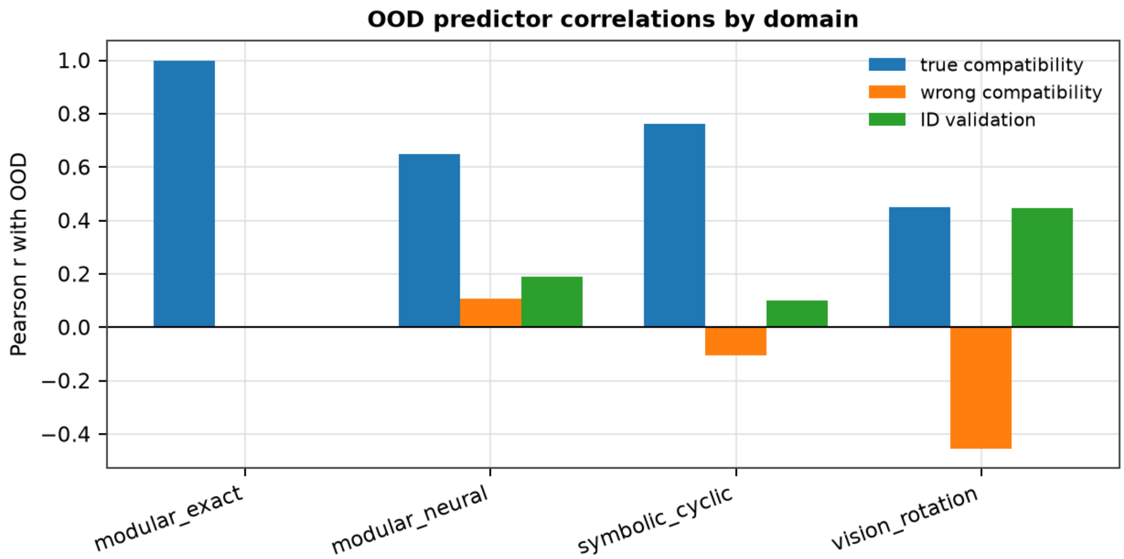

Figure 1. Predictor correlations by domain. True compatibility is compared against wrong-group compatibility and ID validation.

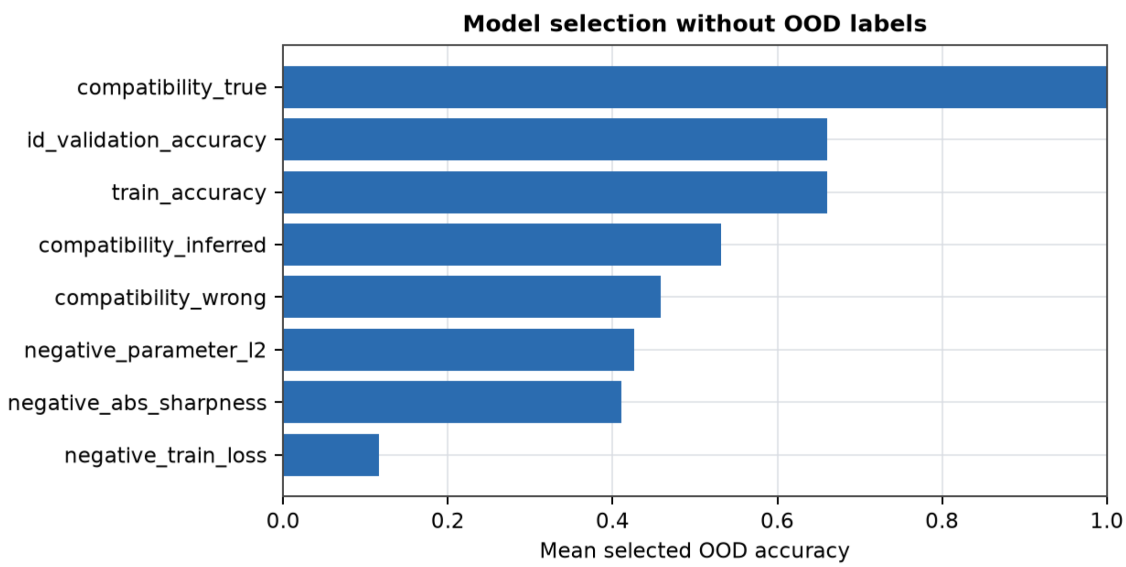

Figure 2. OOD-free model selection. Models are selected by each proxy among train/ID-equivalent candidates and evaluated only after selection.

4. Selection Without OOD Labels

Predictor	Domains	Mean selected OOD
compatibility_true	3	1.000
id_validation_accuracy	3	0.660
train_accuracy	3	0.660
compatibility_inferred	2	0.531
compatibility_wrong	3	0.458
negative_parameter_l2	3	0.426
negative_abs_sharpness	3	0.410

negative_train_loss	3	0.117

Table 3. Selection by proxy among ID-equivalent models.

5. Operating Regime

This is not full OOD certification. The allowed claim is narrower: for finite or structured domains where deployment shift is generated by a candidate transformation family, measured compatibility can predict and improve model selection under underspecification.

The next regime transition is learned or weakly inferred transformation discovery, followed by compatibility-guided regularization and active data acquisition.